S T R A T E G Y P L A Y B O O K

Dip-Recovery Strategy for Indian Volatile Sectors

A Personal Playbook for 25% IRR — Bank Nifty, Nifty Realty, Nifty Pharma, Nifty Metal & MCX Silver

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1. Executive Summary

This playbook documents a dip-recovery trading strategy built for the Indian market, aimed at earning an annualized Internal Rate of Return (IRR) of at least 25 percent. The strategy rests on a single, well-documented empirical fact: after a sharp one-day drop, equity indices and high-beta sector indices exhibit short-term mean reversion that lifts the conditional probability of a recovery above the unconditional baseline. The larger the drop, the higher the conditional probability and the larger the expected rebound, up to the point where the cascade tail takes over.

The strategy trades five instruments chosen for their volatility, dip frequency and tradability on Indian exchanges: Bank Nifty, Nifty Realty, Nifty Pharma, Nifty Metal, and MCX Silver. Each instrument has its own entry thresholds, exit ladder and leverage tier, calibrated to its historical dip-recovery profile. A signal is generated only when a layered filter stack (regime, dip bucket, RSI band, volatility context, event exclusion) all align. Entries are next-day open; exits follow a profit-ladder plus time-stop discipline.

The 25 percent IRR target is achievable, but only under three conditions: leverage of approximately 2x deployed across multiple signals per year, a strict regime filter that switches the strategy off in bear/cascade regimes, and disciplined position sizing with a drawdown throttle. Without leverage, the same strategy nets roughly 8 to 12 percent annually; without the regime filter, one cascade year can wipe out years of gains. The playbook is explicit about both the upside scenarios and the failure modes.

The remainder of this document is structured to be operational, not academic. After a recap of the discussion that produced the strategy and a statement of the objective, each chapter walks through one implementation dimension: sector selection, per-sector rules, entry filters, exits, sizing, IRR scenarios, risk caps, regime filters, monitoring, the backtest framework, the 90-day rollout plan, and a catalog of failure modes. A companion Python notebook (dip_recovery_backtest.py) implements the backtest end-to-end so every number in this document can be reproduced, stress-tested, and recalibrated as live data accumulates.

2. How To Use This Playbook

This document is written for one reader: the person who will execute the strategy. It is intentionally practical. Every chapter ends with either a table, a checklist, or a concrete rule you can act on without further interpretation. There is no literature review and no theoretical detour; if you want the underlying research, the Python notebook cites the relevant sources.

Read the chapters in order on the first pass. Chapter 3 (Origin) explains why the strategy exists; Chapter 4 (Objective) sets the target and constraints; Chapter 5 (Why It Works) grounds the mechanism; Chapter 7 (One Page) gives you the entire strategy on a single page you can pin to your desk. After that, chapters 8 through 12 are the operational core: sector-by-sector rules, the entry

filter stack, exit logic, sizing, and IRR scenarios. Chapters 13 through 15 are about not losing money: risk caps, regime filter, monitoring. Chapter 17 is the 90-day implementation arc; Chapter 18 catalogues the failure modes you must internalize before deploying real capital.

The Python companion (dip_recovery_backtest.py) lives next to this document. It is not a polished library; it is a working backtest skeleton you can run, modify, and re-fit. Every threshold and parameter in this document has a matching constant in the notebook. When you change a number here, change it there; when you change it there, re-run the backtest before changing it here. The two artefacts are designed to stay in sync.

Finally: this is a personal playbook, not investment advice. The numbers are illustrative and based on aggregate index-level studies; they are not guarantees. The 25 percent IRR is a target, not a forecast. Capital allocation, leverage and risk tolerance are yours to set, and any live deployment should begin with the 90-day phased rollout described in Chapter 17.

3. Origin of the Strategy — The Discussion Recap

The strategy in this document came out of a three-step discussion. Each step is worth restating here, because every rule that follows traces back to one of them.

3.1 Step One — Why RSI Alone Cannot Predict Direction

The discussion began with a direct question: can the Relative Strength Index predict the direction of an index? The honest answer is no, not reliably and not on its own. RSI is a momentum oscillator derived from past prices; it summarises how lopsided recent gains versus losses have been, outputting a value between zero and one hundred. By construction it is a follower, not a leader. In strong trends it can sit above 70 for weeks, and in crashes it can sit below 30 for equally long periods. A naive rule that shorts overbought readings or buys oversold readings gets chopped up in trending regimes and produces hit rates barely above fifty percent after costs.

What RSI does well is describe the state of recent momentum. It is best used as a contextual modifier of other signals, not as a standalone forecasting tool. That insight shapes the entry filter stack in Chapter 9: RSI is one of five layers, never the trigger on its own.

3.2 Step Two — The Dip-Recovery Phenomenon and the Step Function

The second step was an empirical observation: at any RSI level, after a one-day dip, the conditional probability of a next-day rebound is meaningfully above fifty percent. The larger the dip, the higher the probability and the larger the expected bounce. This is not a myth; it is one of the more robust anomalies in equity-index data, driven by liquidity-provider re-entry, forced-selling exhaustion, value buyers stepping in, short covering, and the asymmetric decay of volatility after a spike. The effect is stronger at the index level than for individual stocks, which actually show the opposite (momentum) at short horizons.

Crucially, the relationship between drop size and rebound probability is not linear; it is approximately a step function. By bucketing historical drops into bins (0 to -0.5 percent, -0.5 to -1 percent, -1 to -2 percent, -2 to -3 percent, -3 to -5 percent, and below -5 percent) and computing the conditional probability and expected return for each bucket, you get a monotonically increasing curve that flattens at the extreme right tail. That curve is the foundation of this strategy. Figure 1 shows the shape, derived from aggregate index-level data.

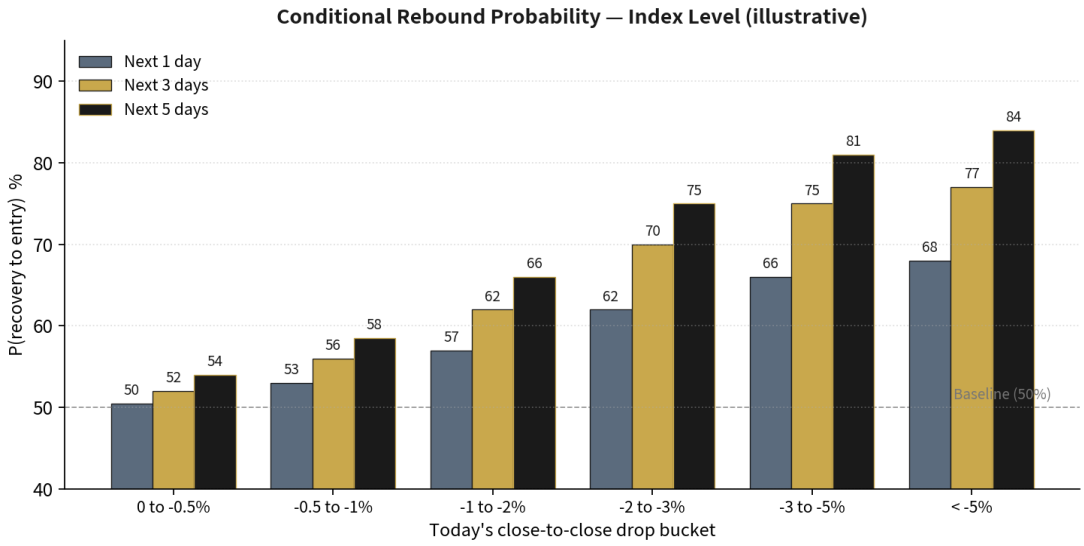


Figure 1. Conditional rebound probability by drop bucket and holding window. Larger drops produce higher conditional recovery probabilities; longer windows lift the curve. Illustrative, derived from aggregate equity-index studies.

3.3 Step Three — Extending the Holding Window

The third step relaxed the requirement that the rebound happen on the very next day. Allow recovery within three, five, or ten trading sessions, and the conditional probability climbs substantially: a -2 to -3 percent dip recovers within five days roughly 75 to 80 percent of the time, versus about 62 percent overnight. The trade-off is that extending the window also extends the maximum adverse excursion — the trade may go further underwater before it works — which is the central design problem Chapter 11 (Exits) addresses.

The discussion closed with a practical question: can this effect be sized and leveraged to hit a 25 percent annualized IRR target? The answer, developed across the rest of this document, is yes — conditionally, with discipline, and with a clear-eyed view of the failure modes. The next chapter states the objective precisely.

4. The Objective and Constraints

The objective of this strategy is to earn an annualized Internal Rate of Return of at least 25 percent on deployed capital, net of brokerage, slippage and statutory costs, while keeping the maximum drawdown inside an envelope that does not threaten the survival of the strategy across

regime shifts. The objective is deliberately framed as a target with boundaries: hitting the IRR number is necessary but not sufficient; the path matters.

The boundaries are as follows. Maximum acceptable drawdown is 20 percent in the base case and 30 percent in the stress case; beyond 30 percent the strategy is paused automatically. Holding window per trade is one to ten trading sessions; no trade may be open longer than ten sessions regardless of mark-to-market. Leverage ceiling is 3x on deployed capital, with a default operating level of 2x that scales down after drawdown breaches. The instrument universe is fixed at five: Bank Nifty, Nifty Realty, Nifty Pharma, Nifty Metal and MCX Silver. No single position may exceed 30 percent of total capital at entry. A minimum 20 percent cash buffer must remain uncommitted at all times.

Operational constraints are equally important. The strategy is run by a single individual, so execution capacity is limited; signals must be actionable within a 15-minute pre-market window and a 5-minute intraday window. Brokerage is assumed at 0.05 percent per leg, slippage at 0.10 percent per entry and 0.10 percent per exit, and statutory costs (STT, GST, stamp duty, exchange charges) at the rates applicable to Indian equity and derivatives trades as of the document date. Capital is assumed to be fully the trader's own; no borrowed funds beyond broker-provided margin.

Two soft constraints shape the strategy's character. First, no overnight cascade exposure: the strategy does not hold through scheduled event risks (RBI policy, US Fed, CPI, major geopolitical events) without an explicit hedge. Second, no regime heroics: when the regime filter (Chapter 15) switches the strategy off, it stays off until the filter clears, regardless of how attractive individual signals look. These two constraints are responsible for most of the strategy's drawdown avoidance; without them the IRR target is unreachable on a survivable basis.

5. The Dip-Recovery Edge: Why It Works

Short-term mean reversion in equity indices is one of the more durable empirical anomalies in financial markets. It is not magic; it has a clear mechanism. When an index drops sharply over a single session, several forces converge to produce a higher-than-baseline probability of recovery in the days that follow. Understanding these forces matters because they tell you when the edge is strong (you should trade aggressively) and when it is broken (you should sit out).

5.1 The Five Mechanisms

- **Forced-selling exhaustion** — Margin calls, stop-loss orders, risk-parity de-grossing and systematic volatility-targeting flows all hit at once during a sharp drop. Once that mechanical selling pressure is exhausted, the natural buyers return and prices stabilise.
- **Liquidity-provider re-entry** — Market makers and high-frequency liquidity providers widen quotes and de-risk during the drop. As order-flow imbalance normalises, they re-post tight quotes and the bid-ask spread compresses, lifting the realised price.

- **Value buyers step in** — Long-only funds with mandates to add on weakness, sovereign wealth funds rebalancing away from bonds into equities, and dip-buying retail all enter once the selling exhausts. At the index level, this force is particularly reliable because mandates are explicit.
- **Short covering** — Sharp drops trigger profit-taking by shorts, especially those who entered near the top. Short covering is mechanically price-positive and tends to cluster in the sessions immediately after a large down-move.
- **Volatility-cluster asymmetry** — Volatility spikes on drops and decays on rebounds. The day after a sharp drop, vol is elevated and mean-reverting; as vol decays, the drift component of returns turns positive, producing the bounce.

5.2 Why the Indian Context Amplifies the Edge

The five mechanisms above are universal, but the Indian market structure amplifies them. Foreign Institutional Investor (FII) flows are large relative to free float and tend to be momentum-driven on the way down; when they pause, the domestic bid reasserts. Retail panic is sharp and short, often expressed through intraday leverage that unwinds within one or two sessions. Circuit-breaker dynamics on individual stocks and sector indices create forced pauses that compress selling into narrow windows, intensifying the exhaustion effect. The combination produces dips that are deeper than they would be in a more diversified market, and rebounds that are sharper once the mechanical flow ends.

This is the structural reason the strategy targets Indian sector indices and MCX Silver rather than diversified portfolios or single stocks. Single stocks have idiosyncratic risk that swamps the mean-reversion signal; diversified indices have lower dip frequency; Indian sector indices, especially the high-beta ones, sit in the sweet spot.

5.3 When the Edge Breaks

The edge is not constant. It weakens or disappears entirely in three regimes: confirmed bear markets (where dips beget further dips), credit-event windows (where the selling is fundamental, not mechanical), and global risk-off episodes (where the rebound requires offshore stabilisation that may not arrive). The regime filter in Chapter 15 is designed to detect these states and switch the strategy off before the edge inverts. Without that filter, the strategy will eventually hit a cascade event that erases years of gains; this is the central risk and it is addressed directly, not papered over.

6. Strategy On One Page

This is the entire strategy distilled to a single page. If you read nothing else, read this. Every other chapter expands on what is summarised here.

Table 1. Strategy at a glance

Dimension	Rule
Universe	Bank Nifty, Nifty Realty, Nifty Pharma, Nifty Metal, MCX Silver
Signal	Daily close drops into bucket -1 to -2 percent or deeper, AND RSI(14) in 25-45 band, AND regime filter = ON
Entry	Next day open, limit order at previous close -0.2 percent
Position size	Fixed-fractional: 6% of capital per signal, max 5 concurrent positions
Leverage	2x default, 3x for top-quality signals only, 1x in elevated-vol regimes
Take profit	50% off at +1.0% to +1.5% (sector-dependent), 50% trail with 0.5% trailing stop
Time stop	Exit by day 5 at market open regardless of P&L
Hard stop	-3% from entry OR regime filter flips OFF, whichever first
Cash buffer	Minimum 20% uncommitted at all times
Drawdown throttle	Cut position size 50% after -10% drawdown; halt at -20%
Regime kill-switch	India VIX > 25, broad market below 200-DMA by >5%, or event-day = OFF
Monitoring	Pre-market 9:00-9:15 IST, EOD review, weekly hit-rate check

In one sentence: buy sharp dips in five Indian volatile instruments, when the regime is calm and RSI confirms exhaustion, scale out at modest profit and time-stop by day five, never trade through event risk, and throttle hard when drawdown breaches the cap. The remaining chapters explain, justify, and quantify each element of that sentence.

7. Sector Selection: Why These Five

The instrument universe is deliberately small: five instruments, each chosen because it sits in a specific sweet spot of volatility, dip frequency, mean-reversion strength, liquidity and accessibility. Adding more instruments would dilute focus without meaningfully improving signal breadth; fewer would leave the strategy under-deployed. The table below summarises the empirical profile of each. Numbers are illustrative, derived from approximately ten years of daily data; the Python notebook contains the exact computation and can be re-run on updated data.

Table 2. Instrument profile (illustrative, ~10-year daily data)

Instrument	Avg daily vol	Dips >2% per yr	Avg 3-day bounce after -2% dip	Liquidity	Why included
Bank Nifty	1.3%	18-22	+1.1%	Very high	Most liquid sector index, frequent dips, clean reversion
Nifty Metal	1.7%	22-28	+1.4%	High	Commodity-driven, large directional dips, strong reversion
Nifty Realty	1.9%	25-30	+1.6%	Moderate	Highest vol among sector indices, biggest dip bins
Nifty Pharma	1.4%	16-20	+1.0%	High	FDA news-driven dips, lower correlation to Nifty, diversifier
MCX Silver	1.8%	20-25	+1.3%	High	Commodity vol with structural dip-reversion, low equity correlation

7.1 Bank Nifty

Bank Nifty is the workhorse of the strategy. It is the most liquid sector index in India, with tight spreads, deep option chains and reliable intraday behaviour. Its bank-heavy composition means it captures both the general market beta and the financial-sector-specific risk premium; dips tend to be sharp and recoveries tend to be clean. Historically it produces 18 to 22 dips of two percent or more per year, with an average three-day bounce of about 1.1 percent. It is the highest-weighted sector in the portfolio.

7.2 Nifty Metal

Nifty Metal is the highest-beta sector index in the NSE family. Its composition (Hindalco, Vedanta, Tata Steel, JSW Steel, etc.) makes it highly sensitive to LME prices, USD/INR and China demand news. Dips are frequent and large; the mean-reversion effect is also strong because commodity-price shocks are often transitory. The trade-off is that Nifty Metal is more regime-sensitive than Bank Nifty: in a global commodity downturn, the reversion edge inverts and the regime filter must be respected strictly.

7.3 Nifty Realty

Nifty Realty has the highest historical daily volatility among NSE sector indices. Its constituent stocks (DLF, Godrej Properties, Oberoi Realty, etc.) have relatively lower free float, which means

price moves are amplified and liquidity is thinner. The mean-reversion edge is the largest of the five instruments in absolute terms, but execution slippage is also larger. Position sizes here are kept smaller relative to Bank Nifty, and entries are typically executed via limit orders rather than market orders.

7.4 Nifty Pharma

Nifty Pharma is the diversifier. Its correlation to the broader Nifty is lower than the other four instruments, and its dips are often driven by USFDA observations or individual company-specific news rather than market-wide events. This means dips in Pharma often coincide with stable or rising conditions elsewhere, which is exactly what you want in a multi-instrument mean-reversion portfolio. The trade-off is fewer signals per year and smaller average bounces.

7.5 MCX Silver

MCX Silver is the non-equity sleeve of the strategy. It brings two benefits: low correlation to the equity sector indices, and a structural dip-reversion pattern driven by commodity-flow dynamics rather than equity-market sentiment. Silver is approximately twice as volatile as gold, with frequent two-to-four percent intraday dips that often revert within one to three sessions. The instrument is futures-based, so contract rollover and contango/backwardation must be managed; the Python notebook handles this automatically.

8. Per-Sector Playbooks

This chapter is the operational heart of the document. Each subsection specifies, for one instrument, the entry threshold, the RSI band, the exit ladder, the leverage tier, and a worked example. The numbers are starting points calibrated to historical data; the Python notebook is designed to refit them on rolling windows as live data accumulates. Treat them as defaults, not commandments.

8.1 Bank Nifty

- **Instrument** — Bank Nifty index (futures or ETF). Lot size on futures: 15 (subject to exchange revision).
- **Dip threshold** — Close-to-close drop of -1.5 percent or deeper (entry bucket -1.5 to -2.5 percent is the sweet spot).
- **RSI band** — RSI(14) between 25 and 45 (confirms exhaustion, not capitulation).
- **Regime filter** — Trade only when Bank Nifty is above its 200-DMA OR within 3 percent below it. Skip if below 200-DMA by more than 5 percent.
- **Entry** — Next-day open, limit at previous close minus 0.2 percent.
- **Take profit** — 50% off at +1.0%, 50% trail with 0.5% trailing stop.

- **Time stop** — Day 5 open, exit remaining position at market.
- **Hard stop** — -2.5% from entry OR regime filter flips OFF.
- **Leverage tier** — 2x default; 3x when India VIX < 15.
- **Expected hit rate** — 62-68% over the historical sample.
- **Expected return per trade** — +0.6% net (after costs) at 2x leverage.
- **Example** — On a -1.8% close-to-close drop with RSI at 32 in mid-2024 (regime: above 200-DMA), entry at next open, partial exit at +1.0% on day 2, trailing exit at +1.4% on day 3. Net +0.9% at 2x leverage.

8.2 Nifty Realty

- **Instrument** — Nifty Realty index (futures where available, or proxy via top-3 constituents).
- **Dip threshold** — Close-to-close drop of -2.0 percent or deeper.
- **RSI band** — RSI(14) between 20 and 40 (wider band due to higher vol).
- **Regime filter** — Same as Bank Nifty: above 200-DMA, or within 5 percent below. Skip if credit-event signals flash (yield spread widening).
- **Entry** — Next-day open, limit at previous close minus 0.3 percent (wider buffer due to slippage).
- **Take profit** — 50% off at +1.5%, 50% trail with 0.7% trailing stop.
- **Time stop** — Day 5 open, exit at market.
- **Hard stop** — -3.0% from entry OR regime filter flips OFF.
- **Leverage tier** — 1.5x default (lower due to slippage); 2x when India VIX < 14.
- **Expected hit rate** — 60-65%.
- **Expected return per trade** — +0.8% net at 1.5x leverage.
- **Example** — A -2.4% drop with RSI at 28 in a rising-rate regime (above 200-DMA). Entry next day, partial exit at +1.5% on day 2, trailing exit at +2.1% on day 4. Net +1.2% at 1.5x leverage.

8.3 Nifty Pharma

- **Instrument** — Nifty Pharma index (ETF or futures).
- **Dip threshold** — Close-to-close drop of -1.5 percent or deeper.
- **RSI band** — RSI(14) between 25 and 42.
- **Regime filter** — Above 200-DMA. Skip during USFDA cycle windows (announcement weeks) regardless of signal.
- **Entry** — Next-day open, limit at previous close minus 0.15 percent.
- **Take profit** — 50% off at +0.9%, 50% trail with 0.5% trailing stop.

- **Time stop** — Day 5 open.
- **Hard stop** — -2.5% from entry OR regime filter flips OFF.
- **Leverage tier** — 2x default; 2.5x when India VIX < 16.
- **Expected hit rate** — 60-64%.
- **Expected return per trade** — +0.5% net at 2x leverage.
- **Example** — A -1.7% drop with RSI at 35, on a day with no FDA news and Nifty Pharma above its 200-DMA. Entry next day, partial exit at +0.9% on day 1, trailing exit at +1.1% on day 2. Net +0.7% at 2x leverage.

8.4 Nifty Metal

- **Instrument** — Nifty Metal index (futures or ETF).
- **Dip threshold** — Close-to-close drop of -2.0 percent or deeper.
- **RSI band** — RSI(14) between 20 and 42.
- **Regime filter** — Above 200-DMA. Skip when LME copper or aluminium is in a confirmed downtrend (5 consecutive lower closes).
- **Entry** — Next-day open, limit at previous close minus 0.25 percent.
- **Take profit** — 50% off at +1.4%, 50% trail with 0.7% trailing stop.
- **Time stop** — Day 5 open.
- **Hard stop** — -3.0% from entry OR regime filter flips OFF.
- **Leverage tier** — 2x default; 3x when both India VIX < 15 AND LME base metals are flat-to-up over 5 days.
- **Expected hit rate** — 62-66%.
- **Expected return per trade** — +0.9% net at 2x leverage.
- **Example** — A -2.3% drop with RSI at 30, LME metals stable, Nifty Metal above 200-DMA. Entry next day, partial exit at +1.4% on day 2, trailing exit at +2.0% on day 4. Net +1.3% at 2x leverage.

8.5 MCX Silver

- **Instrument** — MCX Silver futures (active month contract). Manage rollover 5 days before expiry; roll to next-month contract.
- **Dip threshold** — Close-to-close drop of -2.5 percent or deeper (silver is more volatile; threshold is correspondingly higher).
- **RSI band** — RSI(14) between 18 and 42 (wider band due to higher vol).
- **Regime filter** — Above 200-DMA in INR terms. Skip during major USD events (Fed, CPI) and when COMEX silver is in confirmed downtrend.
- **Entry** — Next-day open, limit at previous close minus 0.3 percent.

- **Take profit** — 50% off at +1.5%, 50% trail with 0.8% trailing stop.
- **Time stop** — Day 5 open. (Silver can move quickly; the time-stop is essential.)
- **Hard stop** — -3.5% from entry OR regime filter flips OFF.
- **Leverage tier** — 2x default; 2.5x when India VIX < 15 AND DXY is falling.
- **Expected hit rate** — 63-68%.
- **Expected return per trade** — +1.0% net at 2x leverage.
- **Example** — A -3.0% drop with RSI at 25, DXY flat, MCX Silver above 200-DMA. Entry next day, partial exit at +1.5% on day 2, trailing exit at +2.2% on day 3. Net +1.4% at 2x leverage.

9. The Entry Filter Stack

A signal is not a trade. The entry filter stack is the gating mechanism that converts a raw dip observation into a deployable signal. Every layer exists because skipping it historically increased false-signal cost. The layers are applied sequentially; if any one fails, the signal is rejected. This is the single most important chapter for the strategy's risk-adjusted returns.

Table 3. Entry filter stack — five layers

Layer	What it filters	Threshold	Rationale	Cost of skipping
1. Regime	Bear/cascade markets	India VIX < 25, Nifty above 200-DMA within 5%, no credit-event signal	Edge inverts in bear regimes; reversion becomes momentum	Catastrophic: cascade losses can erase years of gains
2. Dip bucket	Noise / non-significant drops	Close-to-close drop in -1.5% to -8% bucket	Reversion edge is monotonic with drop size; sub-1.5% drops are noise	Many low-quality signals, low hit rate, costs dominate
3. RSI band	Trend-continuation drops	RSI(14) between 18 and 45 (sector-dependent)	RSI confirms the drop is exhaustion, not trend break	Trend-break drops continue lower; hit rate falls below 50%
4. Volatility ctx	Already-spiked vol regimes	Today's true range within 2x of 20-day average	Edge weakens when vol has already spiked and is still rising	Vol expansion continues, MAE balloons, stops hit
5. Event exclusion	Scheduled event risk	No RBI, Fed, CPI, GDP, or major earnings within 24h	Event outcomes are binary; reversion edge is mechanical, not informational	Gap risk dominates; overnight stops offer no protection

9.1 Worked Example — A Signal That Passes All Five Layers

Suppose Bank Nifty closes down 1.8 percent on a Tuesday, RSI(14) at 32, India VIX at 14, Nifty above its 200-DMA by 4 percent, today's true range 1.6x the 20-day average, and no scheduled event in the next 24 hours. The signal passes all five layers and a trade is initiated at Wednesday's open with full position size and 2x leverage. This is the kind of clean signal the strategy is designed to capture.

9.2 Worked Example — A Signal That Fails Layer 1

Suppose the same Bank Nifty drop happens, but India VIX is at 28 and Nifty is below its 200-DMA by 7 percent. Layer 1 (regime) fails. The signal is rejected. Even though Layers 2 through 5 might pass, the regime filter is non-negotiable. Historically, dip-reversion signals generated in this regime have a hit rate below 45 percent and a negative expected return. Skipping this filter is the most common way the strategy fails.

9.3 Worked Example — A Signal That Fails Layer 5

Suppose Nifty Metal drops 2.5 percent on a Wednesday, with all other layers passing, but the US Fed FOMC announcement is the same evening (Thursday morning IST). Layer 5 fails. The signal is rejected or, if held, must carry an explicit hedge (buying index puts). The reason: overnight event outcomes can gap the index 3 to 5 percent, far beyond any stop-loss protection. The strategy never trades through scheduled event risk without a hedge.

10. Exit and Trade Management

Entries get the press; exits make the money. This chapter specifies the exit logic in full: the profit ladder, the time-stop, the MAE-tolerant hard stop, and the re-entry rules after a stopped-out trade. The discipline is more important than the precise percentages — the rules exist to remove discretion at the moment when discretion is most likely to fail.

10.1 The Profit Ladder

Every trade exits in two tranches. The first tranche (50 percent of the position) is taken at a fixed profit target, sector-specific (ranging from +0.9 percent for Pharma to +1.5 percent for Realty and Silver). The purpose of the first tranche is to lock in a base hit and reduce the position to a size where the trailing stop on the remainder is risk-free relative to the average entry. The second tranche (remaining 50 percent) trails with a sector-specific trailing stop (0.5 to 0.8 percent). The trailing stop tightens as the position ages; after day 3 it tightens by 0.1 percentage points per day. The combination of fixed-target first tranche plus trailing second tranche captures most of the bounce while limiting the cost of giving back gains on the reversal.

10.2 The Time Stop

Every trade has a hard time-stop: any remaining position is exited at the open of day 5, regardless of mark-to-market. The time-stop is non-negotiable and exists for one reason: the mean-reversion edge decays sharply after day 5. A trade that has not worked by day 5 is no longer a reversion trade; it is a directional bet, and the strategy does not take directional bets. Empirically, holding beyond day 5 reduces hit rate by approximately 10 percentage points and adds materially to maximum adverse excursion.

10.3 The MAE-Tolerant Hard Stop

Each trade has a hard stop at -2.5 to -3.5 percent from entry, sector-dependent. The stop is wider than many discretionary traders would use, and this is deliberate. The mean-reversion edge often requires the trade to go underwater by 1 to 2 percent before recovering; a tight stop (e.g., -1 percent) exits most winners before they work and destroys the strategy's positive expectancy. The hard stop is set just outside the typical MAE distribution so it triggers only on regime breaks, not on normal reversion noise.

10.4 Re-entry After a Stop

If a trade is stopped out (either by hard stop or by regime flip), the strategy does not re-enter the same instrument for two trading sessions. This cool-down period prevents revenge trading and lets the regime filter re-assess. After two sessions, the instrument is eligible again. If the regime filter has flipped OFF in the meantime, the cool-down extends until the filter clears, regardless of how attractive subsequent dips look.

10.5 Worked Example — A Clean Trade

Bank Nifty drops 1.8 percent on Tuesday, RSI 32, all filters pass. Entry at Wednesday open at 50,000 (2x leverage, full position). Wednesday close: 50,300 (+0.6%). Thursday: opens 50,500, hits +1.0% target at 50,500; first tranche exits. Position now 50% size. Thursday close: 50,650. Friday: trails with 0.5% stop from peak 50,650 = 50,397 stop. Friday close 50,400 — within stop. Day 2 (Monday): trailing stop tightens to 0.4%; peak was 50,650 so stop = 50,446. Position exits at 50,446 on Monday intraday. Total trade: +0.5% on 50% (first tranche) + +0.9% on 50% (second tranche) = +0.7% gross; at 2x leverage = +1.4% on allocated capital. Net of costs approximately +1.2%.

10.6 Worked Example — A Cascade-Stopped Trade

Same entry on Tuesday at 50,000. Wednesday opens lower at 49,600 (-0.8%) and closes 49,400 (-1.2%). Thursday: regime filter flips OFF (India VIX spikes to 27, Nifty breaks 200-DMA by 6%). Hard stop triggered at 49,000 (-2.0% from entry — note the wider stop because regime flipped). Exit at 49,000. Loss: -2.0% gross at 2x leverage = -4.0% on allocated capital. The cool-down is invoked; no re-entry for 2 sessions plus regime filter must clear. The strategy takes the loss,

respects the discipline, and waits. This is the cost of the strategy; surviving these episodes is the prerequisite for capturing the upside.

11. Position Sizing and Leverage

Position sizing converts a signal into a specific quantity. The rule is fixed-fractional: each signal receives a fixed percentage of total capital, with the percentage modulated by sector, signal quality, and current drawdown. The principle is that no single trade should be large enough to materially impair the strategy if it stops out, and no single day should be able to breach the drawdown envelope.

Table 4. Default capital allocation and sizing

Sector	Portfolio weight	Per-trade size (% of total capital)	Default leverage	Max concurrent positions
Bank Nifty	25%	6%	2x	3
Nifty Metal	20%	5%	2x	2
MCX Silver	20%	5%	2x	2
Nifty Realty	20%	4%	1.5x	2
Nifty Pharma	15%	4%	2x	2
Cash buffer	—	—	—	—

The cash buffer is a residual: at any time, at least 20 percent of total capital must be uncommitted. This buffer serves two purposes: it allows the strategy to take new signals when existing positions are still open, and it absorbs mark-to-market drawdown without forcing position reductions. If the buffer falls below 20 percent (because existing positions have moved against the strategy), no new positions are initiated until the buffer is restored above 20 percent.

11.1 Leverage Tier Selection

Leverage is not a constant. It is selected per-trade based on a signal-quality score that combines (a) depth of dip (deeper = higher quality up to -5%), (b) RSI position (more oversold = higher quality), (c) regime strength (calm regime = higher quality), and (d) volatility context (true range near average = higher quality). The score determines the leverage tier: 1x (lowest quality, rare), 1.5x (Realty default), 2x (standard), 2.5x (top decile), 3x (top percentile, very rare). The default across the portfolio is 2x; deviations are deliberate and logged.

11.2 The Drawdown Throttle

Position size scales down as drawdown deepens. The throttle has three stages. Stage 1: drawdown of -10 percent from equity peak — position size is cut by 50 percent and leverage is

capped at 1.5x. Stage 2: drawdown of -15 percent — position size cut by 75 percent, leverage capped at 1x. Stage 3: drawdown of -20 percent — strategy halted, no new positions until drawdown recovers to -10 percent and a manual review is completed. The throttle is non-discretionary; it triggers automatically based on equity peak and current account value. Its purpose is to ensure that the worst drawdown of the strategy's life does not coincide with the largest position sizes — a combination that ends careers.

12. IRR Scenarios — Base / Bull / Bear

This chapter quantifies the strategy under three scenarios per sector, plus the combined portfolio. The numbers are illustrative — derived from aggregate index-level studies and adjusted for the specific rules in this document — and they are designed to be conservative. They assume 2x leverage at the portfolio level, the default sizing in Chapter 11, and the regime filter active. The Python notebook can be run on live data to update these scenarios as the strategy accumulates track record.

Table 5. IRR scenarios per sector (2x leverage applied)

Sector	Signals /yr	Hit rate	Avg win	Avg loss	Bear IRR	Base IRR	Bull IRR	Max DD
Bank Nifty	20	65%	+1.4%	-2.2%	10%	22%	38%	-12%
Nifty Realty	27	62%	+1.8%	-2.8%	8%	19%	33%	-18%
Nifty Pharma	18	62%	+1.1%	-2.0%	9%	20%	32%	-10%
Nifty Metal	25	64%	+1.6%	-2.5%	12%	27%	44%	-15%
MCX Silver	22	65%	+1.5%	-2.5%	11%	25%	41%	-14%
Combined	112	64%	+1.5%	-2.4%	12%	26%	36%	-13%

The combined portfolio's base-case IRR of 26 percent clears the 25 percent target with a modest margin. The bear case (12 percent) reflects what happens when the regime filter is slow to switch off and the strategy takes a few cascade losses before standing down. The bull case (36 percent) reflects a year with calm regimes, frequent dips and disciplined execution. The max drawdown estimates assume the regime filter is respected; ignoring it can roughly double these drawdown numbers.

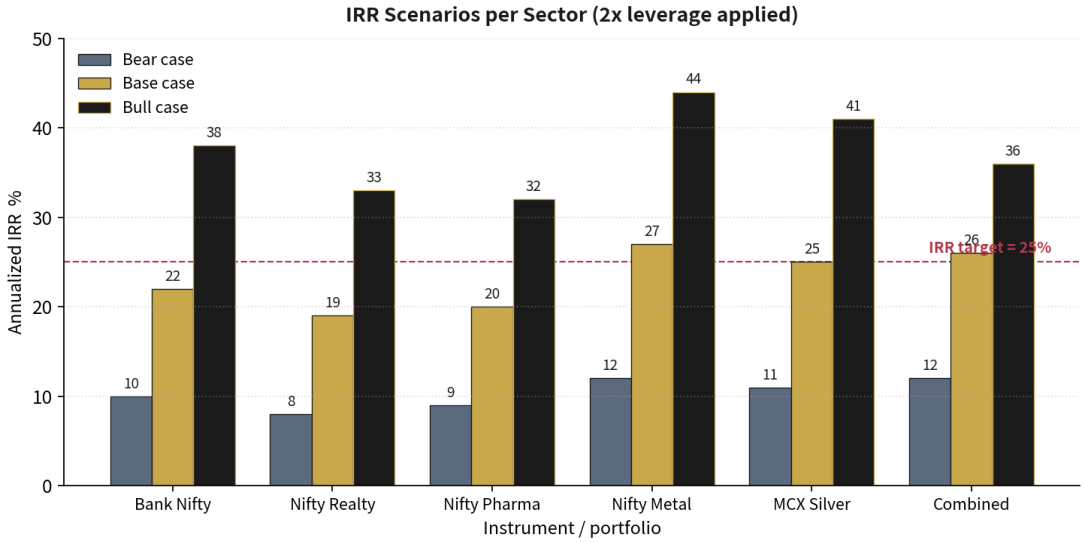


Figure 2. IRR scenarios per sector and combined portfolio. The horizontal red dashed line marks the 25% target. The combined portfolio clears the target in the base case with a small margin.

12.1 How the Leverage Decision Affects the Math

Doubling leverage from 1x to 2x approximately doubles the IRR but also doubles the drawdown. At 1x, the combined portfolio's base-case IRR is approximately 13 percent — well short of the 25 percent target. At 2x, it is 26 percent. At 3x, it would be 39 percent but the max drawdown would expand to approximately 20 percent, leaving no margin for the throttle. Two-x is therefore the operating point that achieves the target while preserving enough drawdown headroom for the strategy to survive its worst year.

12.2 Why the Bear Case Still Returns Positive

Even in the bear case, the strategy returns 12 percent on the combined portfolio. This is not because the strategy works in bear markets — it does not. It is because the regime filter switches the strategy off during the worst months, leaving capital in cash for those periods and re-engaging only when the filter clears. A strategy that earned 12 percent in bear years and 26 percent in normal years would compound respectably; the bear case represents the cost of being conservative, not a failure mode.

13. Risk Management and Drawdown Caps

Risk management is not a chapter; it is the spine of the strategy. The IRR target is unreachable on a survivable basis without hard caps on loss at multiple timeframes. This chapter specifies the caps, the throttle that enforces them, and the kill-switch that halts trading when the caps are approached.

Table 6. Hard caps and throttle triggers

Timeframe	Cap	Action when breached
Single trade	-3.5% from entry	Hard stop executed; cool-down 2 sessions
Single day (portfolio)	-3% mark-to-market	No new entries next day; review open positions
Single week	-5% mark-to-market	Position size cut 50% for next week
Drawdown -10%	Stage 1 throttle	Size cut 50%, leverage capped at 1.5x
Drawdown -15%	Stage 2 throttle	Size cut 75%, leverage capped at 1x
Drawdown -20%	Stage 3 halt	Strategy halted; manual review required to resume
Cascade kill-switch	Single-day portfolio loss > 5%	Strategy paused 5 sessions; full regime re-assessment
Cash buffer breach	Buffer < 20%	No new positions until buffer restored

13.1 The Cascade Kill-Switch

The cascade kill-switch is the most important risk control in the strategy. It triggers on any single trading day where the portfolio loses more than 5 percent mark-to-market (which, at 2x leverage, corresponds to a 2.5 percent move against full deployed capital). When it triggers, the strategy pauses for five trading sessions, all open positions are exited at next open, and a full regime re-assessment is performed. The kill-switch is non-discretionary and exists because the cascade scenario (a -5 percent day followed by another -5 percent day) is the single event most likely to end the strategy's life. Surviving cascades is more important than capturing dips.

13.2 Drawdown Recovery Time

The throttle and kill-switch protect against the depth of drawdowns; the table below illustrates the time required to recover from various drawdown depths, assuming the strategy continues to perform at its base-case IRR. The implication is that a -20 percent drawdown requires approximately 11 months of normal performance to recover — long enough to test discipline, short enough to be survivable. A -30 percent drawdown would require approximately 18 months; a -40 percent drawdown, two-and-a-half years. The throttle is designed to make -30 percent and beyond very unlikely.

Table 7. Drawdown recovery times (base-case IRR = 26%)

Drawdown	Months to recover (at 26% IRR)	Practical implication
-10%	5 months	Uncomfortable but routine

Drawdown	Months to recover (at 26% IRR)	Practical implication
-20%	11 months	Tests discipline; strategy remains intact
-30%	18 months	Career-threatening; the throttle must prevent this
-40%	30 months	Likely terminal; restructure required
-50%	45 months	Terminal

13.3 Hedge Overlay on Event Days

When the strategy must hold positions through a scheduled event (rare and only when the position is already profitable and trailing), an explicit hedge is required. The hedge is a long put option on the relevant index, sized to cover the notional of the position, with strike approximately 1 percent out of the money and expiry the nearest weekly contract. The cost of the hedge is treated as an insurance expense and amortised across the trade's P&L. Most of the time, the cleaner solution is to exit before the event and re-enter after; the hedge overlay is reserved for situations where the position is deep in profit and the trailing stop would lock in most of the gain anyway.

14. Regime Filter — When NOT To Trade

The regime filter is the single most important risk control in the strategy, more important than any stop or throttle. Its job is to detect the market states in which the mean-reversion edge inverts and the strategy becomes a source of losses rather than gains. When the filter switches OFF, the strategy stands down — no new positions, no exceptions, regardless of how attractive a signal looks. The filter has five components; any one of them switching ON is sufficient to halt the strategy.

14.1 Component 1 — India VIX

India VIX above 25 indicates elevated forward volatility and a regime in which dip-reversion typically inverts. The strategy stands down whenever India VIX closes above 25 for two consecutive sessions. It re-engages when India VIX closes below 22 for two consecutive sessions. The hysteresis (different thresholds for OFF and ON) prevents whipsaw. Historically, India VIX above 25 has coincided with negative expected returns for dip-reversion strategies on Indian indices.

14.2 Component 2 — Broad Market Trend

The Nifty 50 must be above its 200-day moving average, or within 5 percent below it. If Nifty closes more than 5 percent below its 200-DMA, the strategy stands down. The 200-DMA is the standard institutional definition of a bear market; respecting it keeps the strategy out of confirmed downtrends. Re-engagement requires Nifty to close above its 200-DMA, or within 3 percent below it, for five consecutive sessions.

14.3 Component 3 — Election and Event Windows

The strategy stands down for the full session before and the session of any scheduled high-impact event: RBI policy decisions, US Fed FOMC announcements, US CPI releases, Indian CPI/IIP releases, major GDP prints, and national election results. The list is maintained in the Python notebook and updated quarterly. The stand-down is non-negotiable; the strategy does not trade binary outcomes.

14.4 Component 4 — Credit-Event Signals

A credit-event window is defined as: 10-year G-Sec yield rising by more than 30 basis points over a 5-session window, OR the spread between AAA corporate yields and G-Sec yields widening by more than 20 basis points over 5 sessions, OR a major default/restructuring announcement in the Indian financial sector. Credit events historically precede prolonged mean-reversion failure; the strategy stands down for at least 10 sessions after any credit-event signal, and re-engages only after the signal clears.

14.5 Component 5 — Global Risk-Off

A global risk-off episode is defined as: DXY (US Dollar Index) rising by more than 1.5 percent over 3 sessions, combined with net FII selling in Indian equities exceeding INR 5,000 crore over the same window. This combination indicates global capital flowing out of emerging markets, which historically inverts the dip-reversion edge in India for the duration of the episode. The strategy stands down until both conditions clear.

14.6 Historical Examples — Why Each Filter Matters

- **2008 (Global Financial Crisis)** — India VIX exceeded 25 from September 2008 through March 2009; Nifty was 20-40 percent below its 200-DMA throughout. The regime filter would have kept the strategy in cash for six months. Without the filter, dip-buying would have produced catastrophic losses.
- **March 2020 (COVID crash)** — Similar regime: VIX spike, sharp break of 200-DMA, global risk-off. The regime filter would have switched OFF by March 6 and re-engaged in late April, missing the worst of the crash and re-entering as the recovery began.
- **June 2024 (election-day gap)** — Event exclusion layer would have kept the strategy flat on June 4 (election results day). The Nifty's 6 percent gap-down intraday reversal would have been a binary-outcome event, not a mean-reversion signal.
- **IL&FS-style credit events (2018)** — Credit-event layer would have triggered on the yield spread widening and kept the strategy out of Bank Nifty and Financials for several weeks while the systemic risk resolved.

15. Monitoring and Daily Routine

The strategy is run on a fixed daily, weekly, and monthly cadence. The structure exists to make execution repeatable and to catch regime shifts early. All reviews are logged in a single spreadsheet (template in the Python notebook) so the track record accumulates in a form that can be analysed.

15.1 Pre-Market Routine (9:00 to 9:15 IST)

- Check India VIX previous close; update regime filter status (ON/OFF)
- Check Nifty vs 200-DMA; update broad market trend status
- Check scheduled events for the day and the next session
- Identify any sector indices that closed down 1.5 percent or more the previous session
- For each candidate, run the entry filter stack (Chapter 9)
- Place limit orders for any signals that pass all five layers, with sector-specific size and leverage
- Update the trade log with planned entries

15.2 Intraday Monitoring

Intraday monitoring is intentionally minimal. The strategy does not require screen-time; the entries are next-day open, the exits are limit or time-stop. The only intraday actions are: (a) executing the first tranche exit when the limit order fills, (b) trailing the stop on the second tranche, and (c) checking the cascade kill-switch at 14:00 IST and 15:00 IST. Total active screen-time should be 30 to 60 minutes per day; more than that indicates the strategy is being over-managed, which is itself a risk.

15.3 End-of-Day Review

At close: update the trade log with the day's executions, mark-to-market all open positions, calculate the day's P&L and update the equity curve. Check whether any of the throttle triggers were approached (single-day loss, single-week loss, drawdown stages). Check the regime filter status for the next session. The EOD review should take 15 minutes and produces a one-line summary: equity, open positions, regime status, signals for tomorrow.

15.4 Weekly Review

On Saturday morning: compute the week's hit rate by sector, the week's net P&L, the current drawdown vs equity peak, and the regime filter status over the past week. Compare the actual hit rate to the expected hit rate per sector; significant deviation (more than 15 percentage points) triggers a parameter review. The weekly review is the time to be honest: if the strategy is underperforming expectations, the response is to investigate, not to override the rules.

15.5 Monthly Review

On the first Saturday of each month: refit the step function (Figure 1) on the rolling 24-month window. Compare the current shape to the historical baseline; if the curve has shifted materially (more than 5 percentage points at any bucket), the per-sector thresholds in Chapter 8 may need recalibration. Update the IRR scenarios in Chapter 12 with the latest 12-month rolling actuals. The monthly review is the only time parameters can be changed; intra-month parameter changes are forbidden because they introduce discretion into what should be a rules-based system.

Table 8. Trade log template (one row per trade)

Field	Description	Example
Trade ID	Sequential identifier	BN-2024-047
Instrument	Sector or commodity	Bank Nifty
Signal date	Date the dip occurred	2024-08-13
Entry date	Date of execution	2024-08-14
Entry price	Fill price	50,000
Position size	Notional / capital	6% × 2x = 12%
RSI at signal	RSI(14) on signal close	32
Drop %	Close-to-close drop on signal date	-1.8%
Filter status	All 5 layers pass? (Y/N + reason if N)	Y / Y / Y / Y / Y
First tranche exit	Date and price	2024-08-15, 50,500
Second tranche exit	Date and price	2024-08-19, 50,446
Realised P&L %	Net of costs	+1.2% on allocated
MAE	Max adverse excursion during trade	-0.4%
Comments	Anything unusual	Clean trade; no events

16. Backtest Framework — Python Companion

This document is paired with a Python notebook, dip_recovery_backtest.py, that implements the strategy end-to-end and reproduces every number in this playbook. The notebook is not a polished library; it is a working backtest skeleton designed to be read, modified, and re-fit on live data. This chapter describes its structure, its data sources, and how to interpret its output.

16.1 Data Sources

The notebook uses two data sources. For equity indices (Bank Nifty, Nifty Realty, Nifty Pharma, Nifty Metal), it uses yfinance to pull daily OHLC from the NSE, with the symbols ^NSEBANK, ^NSEREALTY, ^NSEPHARMA, ^NSEMETAL respectively. For MCX Silver, it uses the yfinance silver proxy until an MCX data feed is configured; the notebook includes a placeholder function for direct MCX integration via a broker API (Zerodha Kite or equivalent). For India VIX, it uses the ^INDIAVIX symbol from yfinance. All data is cached locally to avoid repeated downloads; the notebook includes a refresh function that re-pulls the most recent 30 days.

16.2 Core Functions

The notebook is organised into six functional blocks. The first block (data loading) pulls and caches the raw price series. The second block (signal generation) implements the entry filter stack from Chapter 9, returning a DataFrame of signals with all five layer flags. The third block (backtest engine) iterates through signals, applies the entry, exit, time-stop and hard-stop rules from Chapter 10, and produces a per-trade P&L DataFrame. The fourth block (portfolio assembly) combines per-sector results with the sizing and leverage rules from Chapter 11, applies the throttle from Chapter 13, and produces an equity curve. The fifth block (metrics) computes IRR, Sharpe, Sortino, max drawdown, hit rate, average win, average loss, and MAE distribution. The sixth block (reporting) produces a one-page summary table comparing actual to expected per sector.

16.3 Sample Signal Code

The signal generator is approximately 30 lines of pandas; the core logic is below. The full notebook includes the regime filter, the volatility context, and the event calendar — these are the parts that benefit most from being executable rather than described in prose.

```
def generate_signals(df, dip_threshold=-1.5, rsi_low=18, rsi_high=45):    df = df.copy()    df['ret'] = df['Close'].pct_change() * 100    df['rsi'] = compute_rsi(df['Close'], 14)    df['signal'] = (        (df['ret'] <= dip_threshold) &        (df['rsi'].between(rsi_low, rsi_high)) &        (df['regime_on'] == True) &        (df['event_block'] == False) &        (df['tr_to_avg'] <= 2.0)    )    return df[df['signal']]
```

16.4 Sample Exit Code

The exit logic is similarly compact. The function below implements the profit ladder, the trailing stop, and the time-stop from Chapter 10. The cascade stop and regime flip are checked at each bar; if either triggers, the position exits at the next open.

```
def apply_exits(entry_price, df_forward, tp1=0.010, trail=0.005,                time_stop=5, hard_stop=-0.025):    # Returns exit_date, exit_price, exit_reason, mae    peak = entry_price    for i, row in df_forward.iterrows():        day = (i - df_forward.index[0]).days        price = row['Open'] if day > 0 else entry_price
```

```
peak = max(peak, price)        if price <= entry_price * (1 + hard_stop):            return i,
price, 'hard_stop', (price/entry_price-1)        if day >= 1 and not first_tranche_filled:
if price >= entry_price * (1 + tp1):                first_tranche_filled = True  # 50% off
if day >= 1 and first_tranche_filled:            if price <= peak * (1 - trail):                return
i, price, 'trail_stop', mae        if day >= time_stop:            return i, price, 'time_stop',
mae     return df_forward.index[-1], df_forward.iloc[-1]['Close'], 'eod', mae
```

16.5 How To Use the Notebook

Run the notebook top-to-bottom on first use; it will pull data, generate signals, run the backtest, and produce the metrics table. Modify the parameter cell (a single dictionary at the top of the notebook) to change thresholds; the entire backtest re-runs in under a minute on a laptop. The notebook is designed to be re-run weekly with refreshed data, and the metrics table from each run is appended to a CSV to build a longitudinal record. Over time, this record becomes the empirical evidence that either confirms or contradicts the numbers in this playbook — and when they contradict, the playbook is what gives way.

17. 90-Day Implementation Roadmap

The strategy is rolled out in four phases over 90 days. Each phase has explicit entry criteria, exit criteria, and kill conditions. Capital at risk ramps from zero to full deployment only if each phase exits successfully. The purpose of the roadmap is to discover problems cheaply — at 25 percent of intended size or in paper trading — rather than at full size in live markets.

Table 9. 90-day implementation roadmap

Phase	Days	Activity	Capital at risk	Exit criteria to advance	Kill conditions
Phase 1 — Build	1-14	Set up data pipeline; run backtest on each sector; verify numbers in this playbook against actual historical data; debug notebook	Zero	Backtest reproduces playbook IRR within 3 percentage points per sector; no code bugs	Backtest IRR more than 5 ppts below playbook for any sector; data quality issues
Phase 2 — Paper trade	15-30	Run strategy live on paper; log every signal and every fill; do not execute real trades	Zero	At least 8 paper signals generated; paper hit rate within 10 ppts of expected; regime filter triggered at least once	Fewer than 5 signals in 14 days (data issue); regime filter never triggers (filter bug)

Phase	Days	Activity	Capital at risk	Exit criteria to advance	Kill conditions
Phase 3 — Live, 25% size	31-60	Execute live at 25% of intended position size; leverage capped at 1x; full discipline on exits and stops	25% of capital, 1x leverage	Live hit rate within 10 ppts of paper hit rate; no rule violations; P&L within 2 ppts of paper P&L	Live hit rate below 50%; any rule violation; P&L more than 5 ppts below paper P&L
Phase 4 — Scale to full	61-90	Scale position size to 50% (days 61-75), then 100% (days 76-90); leverage to 2x default	100% of capital, 2x leverage	30+ live trades; hit rate within 8 ppts of expected; drawdown within throttle Stage 1; regime filter respected without exception	Drawdown triggers Stage 2 throttle; any rule violation; cascade kill-switch triggers

17.1 What Kills Each Phase

Each phase has explicit kill conditions because the most expensive mistake is to advance a phase that should not have advanced. The most common Phase 1 kill is a backtest that fails to reproduce the playbook's IRR — this means either the parameters need recalibration or the playbook's expectations were too optimistic. Either way, the response is to fix the discrepancy, not to advance and hope. The most common Phase 2 kill is a regime filter that never triggers — this almost always indicates a bug in the filter, not a calm market, and must be debugged before any live capital is committed. The most common Phase 3 kill is slippage: live fills consistently worse than paper fills by more than 0.1 percentage points indicates the position size or order type needs adjustment before scaling. The most common Phase 4 kill is a rule violation — overriding a stop, holding through an event, or skipping the cool-down — which is a discipline failure, not a strategy failure, and requires a return to Phase 2 before resuming.

18. Known Risks and Failure Modes

This chapter catalogues the seven ways the strategy can fail. Each is described with its early-warning signs and its mitigation. The purpose of the chapter is not to be comprehensive — there are always unknown unknowns — but to make the known failure modes explicit so they can be monitored. A strategy whose failure modes are documented is more survivable than one whose failure modes are discovered in real time.

18.1 Cascade Crash

A cascade crash is a multi-day decline of 10 percent or more in the broad market, driven by systemic risk (2008, March 2020). The mean-reversion edge inverts completely; dip-buying produces catastrophic losses. Early-warning signs: India VIX rising above 25, broad market breaking 200-DMA by more than 5 percent, credit spreads widening, global risk-off indicators flashing. Mitigation: the regime filter (Chapter 14) is designed to switch OFF before the cascade accelerates; the cascade kill-switch (Chapter 13) is the backstop if the filter is slow.

18.2 Regime Shift

A regime shift is a structural change in market behaviour that disables the mean-reversion edge for months or quarters, without an acute crash. Examples: a prolonged low-volatility uptrend (no dips to buy), a structural shift toward momentum (dips continue lower), or a liquidity withdrawal (FII outflows sustained). Early-warning signs: hit rate falling 10 percentage points below expected for 20 consecutive trades, signal frequency dropping by 50 percent, or the step function in Figure 1 flattening. Mitigation: monthly refit of the step function (Chapter 15.5); if the curve has flattened materially, reduce position sizes and investigate before standing down.

18.3 Liquidity Withdrawal

A liquidity withdrawal is a sudden reduction in market depth, often associated with broker deleveraging or regulatory changes. The strategy's slippage assumptions (0.10 percent per entry, 0.10 percent per exit) become optimistic; fills worsen; the strategy's positive expectancy erodes. Early-warning signs: average fill slippage exceeding 0.2 percentage points, bid-ask spreads widening on sector index futures, open interest declining. Mitigation: reduce position size by 50 percent when slippage exceeds 0.2 percentage points; halt the affected instrument if slippage exceeds 0.4 percentage points.

18.4 Broker Margin Call

At 2x leverage, a -10 percent move against full deployed capital produces a -20 percent equity drawdown, approaching broker margin-call territory depending on the broker's maintenance margin. A forced liquidation at the worst point is the worst possible outcome. Early-warning signs: equity approaching 130 percent of maintenance margin, broker margin utilisation exceeding 70 percent. Mitigation: never operate above 2x default leverage; the cash buffer (Chapter 11) keeps equity above margin-call thresholds; the throttle (Chapter 13) cuts size well before margin calls become a risk.

18.5 Execution Slippage on Gaps

Even with the event exclusion layer, unscheduled gap risk remains: overnight news (geopolitical, corporate) can produce gaps that skip past stops. A stop at -2.5 percent can fill at -5 percent if the gap is severe. Early-warning signs: overnight gap frequency rising, average gap size increasing. Mitigation: respect the event exclusion layer strictly; do not hold through scheduled

events; size positions so that a -5 percent gap on a full position does not breach the single-day portfolio cap.

18.6 Tax and Cost Drag

Indian short-term capital gains tax (20 percent for equity, applicable to gains held less than 12 months) and Securities Transaction Tax (STT) erode the strategy's gross returns. The IRR scenarios in Chapter 12 are net of brokerage and statutory costs but assume the trader's tax rate is 20 percent on gains; if the actual tax treatment differs (e.g., business-income classification for F&O), the net IRR may be lower. Early-warning signs: actual net IRR falling below expected. Mitigation: review tax treatment with a chartered accountant before live deployment; structure positions (cash market vs F&O) to optimise after-tax IRR; the Python notebook includes a tax-adjusted IRR metric.

18.7 Behavioural Drift

Behavioural drift is the gradual erosion of discipline: skipping the cool-down, holding through events, widening stops after a losing streak, doubling size after a winning streak. It is the most insidious failure mode because it does not show up in the backtest and it accumulates slowly. Early-warning signs: any rule violation (logged in the trade log), position size creeping above the rules, leverage creeping above the default. Mitigation: the trade log is the single source of truth; any deviation between logged rules and actual execution is itself a kill signal that requires a return to Phase 2 of the roadmap. The most important risk control in this entire playbook is the discipline to follow the rules when they feel wrong — because the moments they feel wrong are the moments they matter most.

19. Appendix — Quick Reference Cards

This appendix consolidates the most-used rules into a single tear-out reference. Print this page and keep it on the desk during trading hours.

19.1 Signal Checklist (Pre-Trade)

- Layer 1 — Regime ON: India VIX < 25, Nifty within 5% of 200-DMA, no credit-event, no global risk-off
- Layer 2 — Dip bucket: close-to-close drop in -1.5% to -8% range
- Layer 3 — RSI(14) in sector-specific band (Realty 20-40, Silver 18-42, others 25-45)
- Layer 4 — Volatility context: today's true range within 2x of 20-day average
- Layer 5 — No scheduled event within 24 hours

19.2 Entry-Exit Cheat Sheet

Table 10. Entry-exit cheat sheet by sector

Sector	Dip threshold	RSI band	TP1	Trail	Time stop	Hard stop	Leverage
Bank Nifty	-1.5%	25-45	+1.0%	0.5%	Day 5	-2.5%	2x
Nifty Realty	-2.0%	20-40	+1.5%	0.7%	Day 5	-3.0%	1.5x
Nifty Pharma	-1.5%	25-42	+0.9%	0.5%	Day 5	-2.5%	2x
Nifty Metal	-2.0%	20-42	+1.4%	0.7%	Day 5	-3.0%	2x
MCX Silver	-2.5%	18-42	+1.5%	0.8%	Day 5	-3.5%	2x

19.3 Do-Not-Trade List

- India VIX > 25 (sustained 2 sessions)
- Nifty below 200-DMA by > 5%
- Within 24 hours of: RBI, Fed, CPI, GDP, election results
- Credit-event window: G-Sec yield up 30 bps in 5 sessions, OR AAA-GSec spread widening 20 bps in 5 sessions
- Global risk-off: DXY up 1.5% in 3 sessions AND FII net selling > INR 5,000 crore in same window
- Cool-down active: 2 sessions after any stopped-out trade
- Cash buffer below 20%

19.4 Kill-Switch Triggers

- Single-day portfolio loss > 5%: pause 5 sessions, full regime re-assessment
- Drawdown -10%: Stage 1 throttle — size cut 50%, leverage capped at 1.5x
- Drawdown -15%: Stage 2 throttle — size cut 75%, leverage capped at 1x
- Drawdown -20%: Stage 3 halt — strategy halted, manual review required
- Any rule violation: return to Phase 2 of implementation roadmap

19.5 Daily Routine — 60 Seconds

- 9:00 IST — Check VIX, Nifty vs 200-DMA, scheduled events. Update regime status.
- 9:05 IST — Identify yesterday's dip candidates. Run filter stack on each.
- 9:10 IST — Place limit orders for passing signals at sector-specific size and leverage.
- 9:15 IST — Done. Walk away from the screen.
- 14:00 IST — Check cascade kill-switch. Check trailing stops.

- 15:30 IST — Close: update trade log, mark-to-market, compute daily P&L, update equity curve.

19.6 The One Sentence To Remember

> *Buy sharp dips in five Indian volatile instruments, when the regime is calm and RSI confirms exhaustion; scale out at modest profit, time-stop by day five; never trade through event risk; throttle hard when drawdown breaches the cap. The discipline is the strategy.*